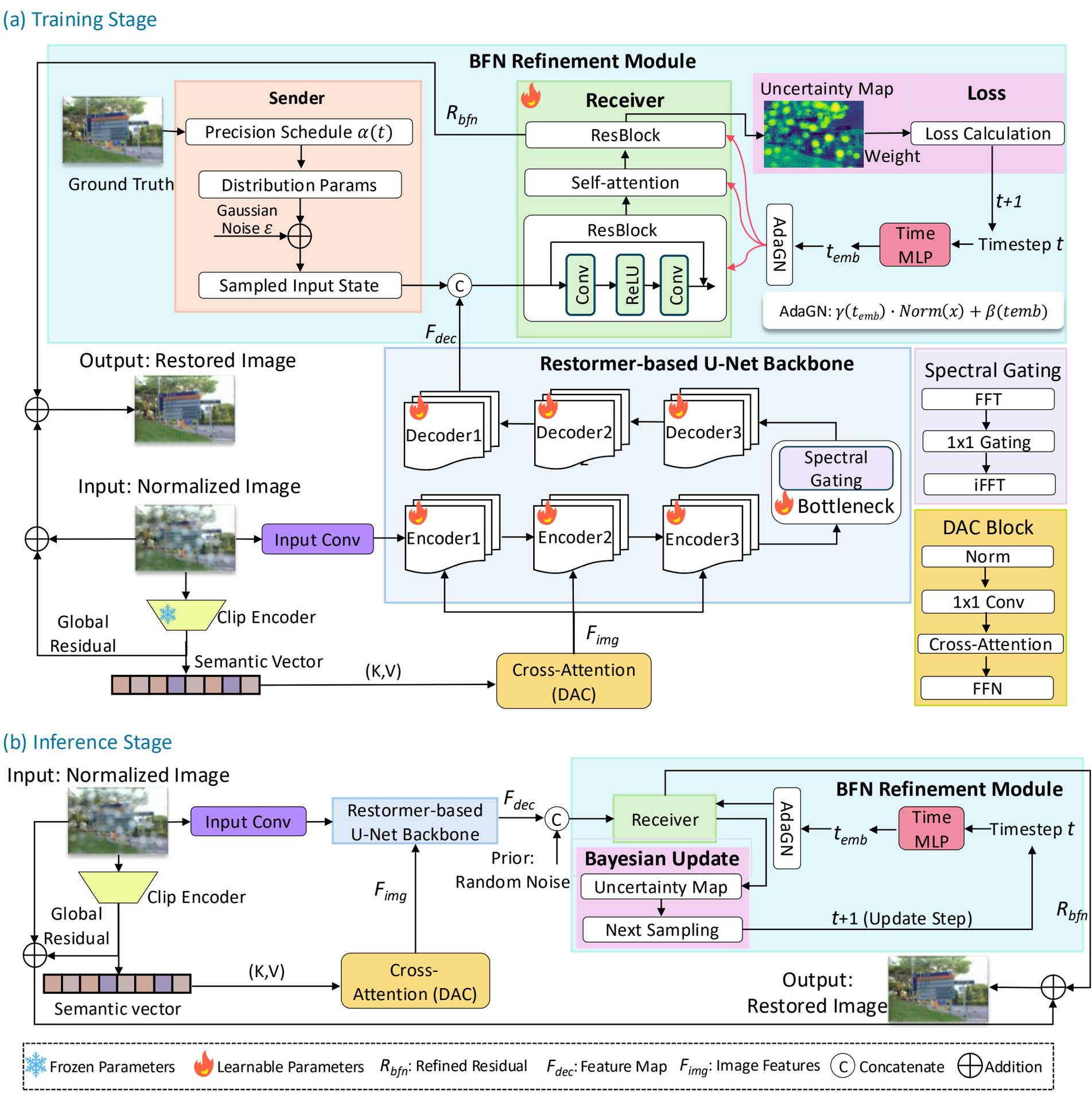
(a) Training Stage
BFN Refinement Module
Sender
Ground Truth
Precision Schedule $\alpha(t)$
Distribution Params
Gaussian Noise ε
Sampled Input State
R_{bfn}
Receiver
ResBlock
Self-attention
ResBlock
Conv
ReLU
Conv
Uncertainty Map
Loss
Loss Calculation
Weight
$t+1$
Timestep t
Time MLP
t_{emb}
AdaGN
AdaGN: $\gamma(t_{emb}) \cdot Norm(x) + \beta(temb)$
F_{dec}
Output: Restored Image
Restormer-based U-Net Backbone
Decoder1
Decoder2
Decoder3
Spectral Gating
Bottleneck
Input: Normalized Image
Input Conv
Encoder1
Encoder2
Encoder3
Spectral Gating
FFT
1x1 Gating
iFFT
DAC Block
Norm
1x1 Conv
Cross-Attention
FFN
Global Residual
Clip Encoder
Semantic Vector
(K,V)
Cross-Attention (DAC)
F_{img}
(b) Inference Stage
Input: Normalized Image
Input Conv
Restormer-based U-Net Backbone
F_{dec}
Prior: Random Noise
BFN Refinement Module
Receiver
AdaGN
t_{emb}
Time MLP
Timestep t
Bayesian Update
Uncertainty Map
Next Sampling
t+1 (Update Step)
R_{bfn}
Clip Encoder
Global Residual
F_{img}
(K,V)
Cross-Attention (DAC)
Semantic vector
Output: Restored Image
Frozen Parameters
Learnable Parameters
R_{bfn}: Refined Residual
F_{dec}: Feature Map
F_{img}: Image Features
Concatenate
Addition